Bequests and late-life housing exit: options, overnight evidence, and the next calibration strategy

Fertility–Housing project

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Executive conclusion

The overnight battery answers a narrower question than "do households have bequest motives?" It asks whether a parent-gated bequest motive becomes useful after adding one particular late-life housing mechanism: a linear decline in the owner's borrowing limit from age 66 onward. The answer is no. The borrowing rule improves the scalar loss only by producing a discrete ownership collapse between ages 74 and 78, and the valid nested comparison puts the bequest strength back at zero. This rejects the borrowing rule and the current combination; it does not reject bequests in a model with a credible old-age block.

Three additional findings determine the next step. First, the age-82 model state forces terminal liquidation, so it cannot be compared with the ACS 82–84 ownership bin. Removing that endpoint does not rescue the borrowing rule: its 74–78 cliff remains. Second, the 12-parameter headline search failed to recover its own 11-parameter submodel, exposing a search-design failure. Third, the fresh Jacobian has rank 9 at a relative singular-value tolerance of 10^{-2} and rank 12 only at 10^{-3}, so the headline system is locally weakly identified even though the θ_0 column is numerically above the repeat-noise floor.

The recommended sequence is therefore:

1. keep the no-bequest clean frontier as the interim benchmark;
2. extend the old-age horizon and add externally measured survival before any new bequest calibration;
3. construct HRS/PSID profiles for ownership transitions, downsizing, mortgage balances, and nonhousing-wealth decumulation;
4. use a smooth, data-pinned owner-exit hazard as a low-cost diagnostic, then decide from those facts whether the structural model needs medical expense risk, a moving/downsizing margin, or a mortgage-balance state;
5. only after an old-age mechanism passes the ownership and wealth gates, reintroduce one free bequest-strength parameter with the restricted winner injected into every expanded search.

1 The decisions and the available options

There are four distinct decisions. Combining them into one calibration obscures which margin failed: the form of estate utility, the source of late-life housing exit, the empirical objects used to discipline that exit, and the optimizer used to compare nested models.

1.1 Estate-utility options

Option B0: no operative bequest motive. The clean frontier fixes $\theta_0 = \theta_n = 0$ and deterministic tenure, leaving 11 estimated coordinates. This is the identified and reproducible interim benchmark. Its limitation is economic rather than numerical: old-age ownership remains too high and the model cannot study estate motives or transfers. Status: retain as the benchmark, not as the final old-age model.

Option B1: child-blind luxury warm glow. The structural savings literature commonly uses a luxury shifter without a child-count multiplier (De Nardi, 2004; Lockwood, 2018; Nakajima and Telyukova, 2017). In the normalization used by the code, the payoff is:

$$v(b) = \theta_0 \left[\frac{(b+\theta_1)^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma} \right]. \tag{1}$$

The subtraction makes zero estate deliver zero utility and prevents a utility level constant from contaminating fertility comparisons. This is the cleanest literature baseline once the old-age block is credible. The published Nakajima and Telyukova (2017) calibration reports an annual shifter of $7,619 in 2000 dollars; there is no verified one-to-one conversion from that number to this model's income normalization, so θ_1 must remain an explicitly external scale until a model-specific estate moment is added.

Option B2: parent-gated luxury warm glow. The implemented headline variant gives the normalized payoff only to parents:

$$v(b, n) = \mathbf{1}\{n \geq 1\}\, \theta_0 \left[\frac{(b+\theta_1)^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma} \right]. \tag{2}$$

This captures the extensive parent-childless contrast used by Hurd (1989) and the modest difference in motive incidence documented by Kopczuk and Lupton (2007). It is less standard than Option B1 because childless households also leave intended estates in the data. Status: diagnostic as an extensive-margin sensitivity, not yet the preferred baseline.

Option B3: equal-division robustness. If an estate is divided equally across n children, a disciplined alternative is:

$$v(b, n) = \mathbf{1}\{n \geq 1\}\, \theta_0 n \left[\frac{(b/n+\theta_1)^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma} \right]. \tag{3}$$

The form follows the equal-division evidence in McGarry (1999) and Wilhelm (1996); it is not a claim that the motive grows linearly with the number of children. Overnight it returned $\theta_0 = 0.0106$ and failed the same ownership test as the headline. Status: diagnostic only.

Option B4: freely estimate both (θ_0, θ_1). This is attractive economically but not identified by the current target system. The overnight A5 arm has 13 free parameters, reached $(\theta_0, \theta_1) = (0.0060, 0.1270)$, and failed the ownership rule. Its low loss is not an estimate because the Jacobian and earlier audits show weak directions. Status: reject as a calibration, retain only for ridge mapping.

1.2 Old-age housing and wealth options

Option E0: the tested borrowing-limit taper. The overnight rule multiplies the owner's collateral share by a deterministic age schedule. It is not a mortgage contract: there is no loan-balance state, origination date, term, refinancing choice, or scheduled payment. Because the constraint tightens sharply while the household is still alive, the rule forces households out of ownership rather than generating gradual repayment. Status: reject.

Option E1: externally measured survival over a longer horizon. Certain death and forced liquidation at age 82 create an artificial endpoint. Extending the horizon to at least the early 90s and loading age-specific survival probabilities is the smallest structural correction. It creates accidental estates and makes the timing of death uncertain, as in Hurd (1989). Survival alone may not generate enough downsizing, but it is a prerequisite for interpreting any bequest motive. Status: recommended first structural change; no calibrated parameter is required.

Option E2: smooth owner-exit or downsizing hazard. An age- and state-dependent owner-to-renter or smaller-home hazard can be pinned from HRS transitions and used as a diagnostic envelope. It would answer whether a smooth exit path is sufficient before a costly state-space expansion. It is reduced form and therefore not a final household problem, but it is far safer than selecting an LTV endpoint by loss. Status: recommended diagnostic after the data construction.

Option E3: medical expense and health risk. Late-life expense risk can generate nonhousing-wealth decumulation and housing exit while preserving endogenous choice. It is standard in structural retirement models and interacts meaningfully with intended versus accidental estates. It requires health states, transition probabilities, and expense distributions, and it may still leave housing too persistent if moving is costly. Status: leading structural candidate if the data show health shocks predict exit.

Option E4: a true mortgage-balance state. If the HRS shows that mortgage repayment, refinancing, or equity extraction is the relevant margin, the model needs a loan balance and contract law of motion. This is the most faithful household-finance option and the most expensive: it adds a state and requires origination, term, interest, payment, and refinancing rules. Reverse-mortgage access can be a later robustness, following Nakajima and Telyukova (2017), but its low incidence makes it a poor default explanation. Status: conditional on the data, not the next automatic build.

Option E5: inter vivos down-payment transfers. The strongest child-directed evidence concerns transfers while parents are alive (Kværner, 2023; McGarry, 1999). A transfer at family formation would relax the model's down-payment constraint directly and may be more relevant to fertility than the terminal estate. This is a separate young-household channel; it does not repair the late-life ownership block. Status: promising subsequent extension, identified with gift incidence and amounts rather than estate moments.

2 What the overnight battery established

2.1 Design

Every arm re-estimated the 11 clean-frontier coordinates against all 15 active moments. A1, A3, and A4 added free θ_0; A5 also freed θ_1 and was labeled unidentified. The owner borrowing endpoint and the headline θ_1 scale were external variants, never selected by loss. The 22 main chains completed 14,711 search evaluations, and the 10 fixed-θ_0 profile chains completed 6,143. Every reported point was solved twice at the tight evaluator with zero repeat difference. Six main-chain and three profile-chain search winners failed the final market-residual gate and were excluded.

2.2 Arm comparison

Lower loss indicates a closer match to the weighted moments, but it is not an acceptance criterion by itself. Table 1 reports tight results; the ownership gate is recomputed over ages 62–78 because age 82 is a forced terminal-liquidation state.

arm	free	loss	θ_0	$\bar{h}_0$	62–78 band	62–78 cliff
A0: control	11	6.7881	0	0.3963	fail	no
A1: parent gate only	12	6.7430	0.0211	0.3638	fail	no
A2: exit only, $\bar{L} = 0.2$	11	4.2024	0	0.2851	fail	yes
A2: exit only, $\bar{L} = 0.4$	11	5.0115	0	0.3683	fail	yes
A2: exit only, $\bar{L} = 0.6$	11	6.1230	0	0.3818	fail	yes
A3: exit + gate, $\bar{L} = 0.4$	12	5.2935	0.0151	0.4237	fail	yes
A3: $\theta_1 = 0.50$ sensitivity	12	5.6877	0.0003	0.3783	fail	yes
A4: equal division	12	5.6473	0.0106	0.3512	fail	yes
A5: both bequest terms free	13	4.9850	0.0060	0.4148	fail	yes

Table 1: Tight arm results. $\bar{L}$ denotes the externally imposed terminal multiplier on the owner borrowing limit, not an observed loan-to-value ratio. A3 at $\bar{L} = 0.2$ produced no tight-eligible chain winner.

The scalar gain comes from the wrong lifecycle path. Under primary A2, ownership at ages 70, 74, and 78 is 0.902, 0.815, and 0.043; under primary A3 it is 0.861, 0.755, and 0.042. The empirical ownership rate is roughly three-quarters throughout these ages. The schedule therefore does not approximate gradual repayment or smooth downsizing.

2.3 The falsifiable bequest prediction

A3 contains A2 exactly when $\theta_0 = 0$. Consequently, a correctly optimized A3 must weakly improve on the A2 loss of 5.0115. The free A3 chain instead reported 5.2935, proving that it missed the known nested basin. The dedicated profile made the same mistake at zero, so the correct profile replaces that cell with the already verified A2 envelope:

fixed θ_0	0	0.05	0.10	0.20	0.40
direct profile loss	8.8134	5.0155	6.1220	8.8669	10.7405
valid envelope loss	5.0115	5.0155	6.1220	8.8669	10.7405

Table 2: Nuisance-reoptimized θ_0 profile. The zero envelope is the algebraically equivalent A2 calibration and must be honored in any nested comparison.

Zero weakly dominates the positive grid. The difference between zero and 0.05 is only 0.004 loss points, smaller than the uncertainty implied by the incomplete basin search. The defensible conclusion is therefore "no evidence for positive θ_0 under this exit rule," not a precise estimate of zero and not a general rejection of bequests.

2.4 Identification

The fresh 15-by-12 Jacobian is deterministic, and the θ_0 column has target-scaled norm 15.70 against a zero repeat-noise floor. Nevertheless, the weighted singular values imply rank 9 at relative tolerance 10^{-2} and rank 12 only at 10^{-3}. Thus three parameter combinations are weak at the more conservative threshold. Since the Jacobian is evaluated at a rejected and non-optimal A3 point, it is a warning rather than an inference result. Freeing θ_1 would worsen the problem; A5 remains unidentified by construction.

3 Plan and decision gates

The next work should be sequential. Each stage has an observable deliverable and a stopping rule, so another large calibration cannot conceal a failed mechanism or optimizer.

stage	action	deliverable	gate
0	Correct the experiment infrastructure.	Exclude the forced terminal age from empirical acceptance unless the horizon is extended; require every expanded optimizer to evaluate and retain the restricted-model incumbent; seed the zero-profile cell from the A2 winner.	A nested model may never report a worse best loss than its restricted submodel.
1	Build the old-age empirical packet before changing the model.	HRS/PSID profiles with provenance and uncertainty for owner-to-renter transitions, downsizing, mortgage balances or LTV conditional on ownership, nonhousing wealth, and the 75+/62–74 decumulation ratio.	No new hard target or borrowing endpoint without a measured counterpart.
2	Extend the horizon and add age-specific survival.	A no-bequest solution through at least the early 90s with the standard diagnostic plots and no forced endpoint inside the empirical window.	Ownership and wealth paths must remain numerically stable at the oldest states.
3	Run a small mechanism frontier with $\theta_0 = 0$.	Compare a smooth data-pinned exit hazard with survival plus health/expense risk. Build a true mortgage state only if the empirical packet points to mortgage balances or refinancing as the relevant source.	Every four-year nonterminal ownership bin lies in the predeclared band; no adjacent decline exceeds 15 points; the wealth path moves in the empirical direction.

stage	action	deliverable	gate
4	Calibrate bequest strength only inside a mechanism that passes Stage 3.	Estimate the 11 core coordinates plus θ_0, keep θ_1 external, inject the Stage-3 winner at $\theta_0 = 0$, and compute a nuisance-reoptimized profile and fresh Jacobian.	The expanded search recovers the zero incumbent; positive θ_0 must improve the tight loss materially, pass the lifecycle gates, and be identified at the declared rank threshold. Otherwise fix $\theta_0 = 0$.
5	Revisit the child channel.	Compare child-blind and parent-gated luxury warm glow; keep equal division as robustness. Add inter vivos down-payment transfers only with gift-incidence and amount moments.	Do not infer a child channel from terminal estates when the identifying variation belongs to transfers.

4 Recommendation requiring author approval

The immediate recommendation is to approve Stages 0–2 and defer a new global calibration. Concretely, keep $\theta_0 = \theta_n = 0$ in the interim benchmark, retire the linear borrowing-limit taper, extend the old-age horizon, and build the HRS/PSID transition and balance profiles. Then run the smooth exit-hazard diagnostic and survival/expense alternative at $\theta_0 = 0$. A full mortgage-balance state should be authorized only if the empirical packet says the mortgage channel is quantitatively central.

Once one old-age mechanism passes its own lifecycle gates, the preferred bequest sequence is Option B1 as the literature baseline, Option B2 as the parent-childless sensitivity, and Option B3 as equal-division robustness. Keep θ_1 externally fixed until an additional estate-distribution or bequest-incidence moment identifies it. Inter vivos transfers are a separate, promising fertility mechanism, but they should follow rather than substitute for repair of the old-age block.

A Complete primary target fits

Tables 4 and 5 report every active target, model moment, gap, weight, and loss contribution. They are tight-evaluator records; the contribution column sums to the reported scalar loss.

Table 4: Primary A2: exit rule only, $\bar{L} = 0.4$, tight loss 5.0115.

moment	target	model	gap	weight	contribution
TFR	1.9180	1.9804	+0.0624	20.0	0.0778
Childless share	0.1880	0.1966	+0.0086	20.0	0.0015
Ownership, ages 30–55	0.5755	0.6443	+0.0688	100.0	0.4733
Parent ownership gap	0.1677	0.1774	+0.0097	45.0	0.0043
Housing increment, 0 to 1 child	0.6644	0.6424	-0.0220	14.0	0.0068
Old parent-childless nonhousing-wealth gap	1.0074	0.8928	-0.1147	2.0	0.0263
Childless renter mean rooms	3.8053	3.8466	+0.0413	6.0	0.0102
Childless owner share with at least 6 rooms	0.5961	0.7645	+0.1684	25.0	0.7090
Old nonhousing-wealth median	2.2305	1.6781	-0.5523	0.8	0.2440
Young childless renter liquid wealth/income	0.1792	0.1808	+0.0016	12.0	0.0000
Childless owner-renter mean-room gap	2.4188	2.2787	-0.1401	12.0	0.2355
Old-age ownership	0.7643	0.8750	+0.1107	160.0	1.9624
Ownership, ages 25–34	0.3412	0.2630	-0.0782	80.0	0.4891
Large-family room gap	0.3677	0.4740	+0.1063	8.0	0.0904
Mean occupied rooms	5.7800	5.4431	-0.3369	6.0	0.6809

Table 5: Primary A3: exit rule plus parent-gated bequests, $\bar{L} = 0.4$, tight loss 5.2935.

moment	target	model	gap	weight	contribution
TFR	1.9180	2.0048	+0.0868	20.0	0.1505
Childless share	0.1880	0.1845	-0.0035	20.0	0.0002
Ownership, ages 30–55	0.5755	0.6047	+0.0292	100.0	0.0854
Parent ownership gap	0.1677	0.1424	-0.0252	45.0	0.0287
Housing increment, 0 to 1 child	0.6644	0.5323	-0.1322	14.0	0.2445
Old parent-childless nonhousing-wealth gap	1.0074	0.9091	-0.0984	2.0	0.0194
Childless renter mean rooms	3.8053	4.0932	+0.2879	6.0	0.4974
Childless owner share with at least 6 rooms	0.5961	0.8208	+0.2247	25.0	1.2623
Old nonhousing-wealth median	2.2305	2.0029	-0.2275	0.8	0.0414
Young childless renter liquid wealth/income	0.1792	0.0888	-0.0904	12.0	0.0981
Childless owner-renter mean-room gap	2.4188	2.1723	-0.2465	12.0	0.7290
Old-age ownership	0.7643	0.8282	+0.0640	160.0	0.6544
Ownership, ages 25–34	0.3412	0.2362	-0.1050	80.0	0.8816
Large-family room gap	0.3677	0.3432	-0.0245	8.0	0.0048
Mean occupied rooms	5.7800	5.4648	-0.3151	6.0	0.5959

B Primary A3 parameter table

The headline arm estimates 12 coordinates and fixes three by external restriction. Only θ_0 is flagged near its search boundary.

parameter	estimate	lower	upper	role
β (annual)	0.983627	0.800	0.9995	estimated
α	0.607209	0.020	0.980	estimated
$\bar{c}_0$	1.19964	0	2	estimated

parameter	estimate	lower	upper	role
$\bar{c}_n$	0.376812	0	3	estimated
$\bar{h}_0$	0.423707	0.05	5.8	estimated
$\bar{h}_{\text{jump}}$	1.41843	0	8	estimated
$\bar{h}_n$	0.119675	0	5	estimated
ψ_{child}	0.078773	-3	3	estimated
κ_{fert}	1.83831	0.02	50	estimated
χ	1.09210	0.1	5	estimated
H_0	8.06146	0.2	80	estimated
θ_0	0.015074	0	1.5	estimated; near zero
κ_{tenure}	0	external		fixed
θ_1	0.25	external scale		fixed
θ_n	0	external		fixed

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